

Autodesk® Scaleform®

Scaleform 4.3

Scaleform 4.3

: Matthew Doyle
: 2.10
: 2013 4 24

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Autodesk® Scaleform® 4.3

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PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

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1 환영

Autodesk® Scaleform® 4.3 . Scaleform® 4.3 ,
SDK ,
 . Scaleform 3.1 , ()
 . , Scaleform Common Lightweight Interface Kit (CLIK™) Video
 .

Scaleform 4.3 :

Scaleform CLIK –
Scaleform Analyzer for Memory and Performance (AMP™) –
ActionScript 2 ActionScript 3
 , 2.5D
Scaleform Kits - UI ()
GFXExport –

CRI Movie™ Scaleform Video
Scaleform Input Method Editor (IME)

2 설치 및 사용

Scaleform 4.3 SDK Scaleform Video, Scaleform IME

<http://gameware.autodesk.com/scaleform/developer/?action=dl>

2.1 *Flash*

Scaleform 4.3 Adobe® Flash® Creative Suite® 3 (CS3) . Scaleform 4.2 Flash
(AS) . Scaleform 4.3 Flash Player 10.1

. Scaleform 4.3 AS2.0 AS3.0 . Scaleform 4.3 Flash
2.0 *Scaleform 4.3– Flash Support Overview*

Creative Suite(CS)

CS

CS4

CS3

Flash, ,

2.2 *SDK*

Scaleform 4.3 SDK

C:\Program Files (x86)\Scaleform\GFx SDK

4.3

Scaleform 4.2

Scaleform 4.3 SDK

SDK

readme_make.txt

readme_xx.txt

DirectX® SDK

, DirectX SDK Scaleform 4.3

Player

Scaleform 4.3 Player DirectX

SDK

Scaleform →

GFx SDK 4.3 → Documentation

C:\Program Files (x86)\Scaleform\GFx SDK 4.3\Doc

2.2.1 설치 패키지

Scaleform

1.

2.

3.

Scaleform

SDK

Scaleform IME

Scaleform

(

).

Scaleform

(

)

SDK

GFXSDK

. Scaleform

.tar.gz

.zip

Scaleform

.

Scaleform

make

. Scaleform SDK

FxPlayer, PlayerSWFToTexture

()

,

Visual Studio

전처리장치 심볼	라이브러리
GFX_ENABLE_VIDEO	Libgfxvideo
GFX_ENABLE_IME	Libgfxime

Scaleform

Shipping

Release

Debug

DebugOpt

2.2.1.1 Windows OS 용 패키지 추출

tar.bz2

Windows

7-zip

tar.bz2

2.2.2 SDK 브라우저

Scaleform 4.3 SDK ,
SDK (-> Scaleform -> GfX 4.3->Scaleform SDK
Browser)
scaleform_gfx_4.3_macos/Bin/Browser/ Browser.app SDK Browser

2.2.3 Scaleform CLIK 설치

Scaleform 4.3 Scaleform Common Lightweight Interface Kit (CLIK) , CLIK
(UI) . SDK

CLIK Flash
. CLIK *Getting Started with CLIK* .

1.

→ → Scaleform → GfX SDK 4.3 → Documentation → Scaleform 4.3 – Getting Started
with CLIK.pdf

2.

C:/Program Files/Scaleform/GfX SDK 4.3/Doc/sf_4.3_getting_started_with_CLIK.pdf

3. SDK

CLIK Scaleform SDK .

:

C:/Program Files/Scaleform/GfX SDK 4.3/Resources/AS2/CLIK/

C:/Program Files/Scaleform/GfX SDK 4.3/Resources/AS3/CLIK/

C:/Program Files/Scaleform/GfX SDK 4.3/Resources/Tools/

:

scaleform_gfx_4.3_platform/Resources/AS2/CLIK/

scaleform_gfx_4.3_platform/Resources/AS3/CLIK/

scaleform_gfx_4.3_platform/Resources/Tools/

2.2.4 Scaleform Video 설치

CRI Movie Scaleform Video (Scaleform)

Scaleform 4.3 SDK . Scaleform Flash

UI Flash .

Scaleform "PlayStation® 3"(PS3™), Xbox® 360, Wii™, Mac PC .

Scaleform Video Scaleform → GfX SDK 4.3 → Video

. Scaleform Video *Getting Started with Video*

.

Video : *C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Tools/VideoEncoder/*

Video : *C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Data/AS2/Video/*

Video :

1. *C:/Program Files/Scaleform/GfX SDK 4.3/Doc/sf_4.3_getting_started_with_video.pdf*

2. → → Scaleform → GfX SDK 4.3 → Documentation → Scaleform 4.3 – Getting Started with Video.pdf

3. SDK Browser.

2.2.5 Scaleform IME 설치

Scaleform (IME) Flash .

PC .

Scaleform 4.3 SDK

C:/Program Files/Scaleform/GFx SDK 4.3/Bin/Data/AS2/IME/

C:/Program Files/Scaleform/GFx SDK 4.3/Bin/Data/AS3/IME/

2.2.6 Platform SDKs

Scaleform

SDK

Scaleform

SDK

Scaleform

[Build](#)

[History](#)

2.2.7 .Net 프레임워크

Scaleform Video CRI movie

NET Framework 2.0 SP1

[Microsoft .NET Framework Version 2.0 SP1](#)

2.2.8 경로 구조

Scaleform

, , , , ,

Scaleform

3rdParty/

Apps/

Samples

Common

FxPlayer

Scaleform Flash player.

GFxPlayerTiny

Scaleform

Bin/

(,)

[Platform]	(ScaleformPlayer,	)
Data/AS2/Samples	.	
FxPlayer	ScaleformPlayer	
RenderTexture	.	
SWFToTexture	.	
ImageDelegate	ImageDelegate	Data/AS2/Video Scaleform
.		
Data/AS2/IME	Scaleform IME sample files	Data/AS2/Kits 2
Data/AS3/Samples	3	
Data/AS3/IME	Sample IME ActionScript 3 files	
Data/AS3/Kits	3	
Doc/		
Include/	Scaleform (C++)	
Lib/	Scaleform	
[Platform]		
[Config]	.	
Obj/	.	
[Platform]	.	
Projects/	.(, .)	
Common	.	
[Platform]	.	
Src/	Scaleform	
GFx	Scaleform SDK	
Render	(.)	
Sound	(.)	

Scaleform

#define

Scaleform 플랫폼	플랫폼 이름	#defines
Win32	Windows	SF_OS_WIN32
x64	Windows for x86_64	SF_OS_WIN32
*-linux	Linux for * processor	SF_OS_LINUX
MacOS	MacOS X	SF_OS_DARWIN, SF_OS_MAC
Android	Android	SF_OS_ANDROID
iPhone	iPhone	SF_OS_IPHONE
PS3	Playstation 3	SF_OS_PS3
NGP	NGP	SF_OS_NGP
3DS	3DS	SF_OS_3DS
Wii	Wii	SF_OS_WII
Xbox360	Xbox 360	SF_OS_XBOX360

	Standard, often gcc
Msvc80	Visual Studio 8.0 (2005)
Msvc90	Visual Studio 9.0 (2008)
Msvc10	Visual Studio 10.0 (2010)
CW	Codewarrior
snc	SN Systems

[Platform]

Win32/Msvc80

"standard"

MacOS
"standard"

2.2.9 Scaleform 구성

Scaleform Visual Studio Visual Studio
IDE make 'Projects'

2.2.9.1 make 사용

Visual Studio IDE 'make' Scaleform
SDK make *readme_make.txt*

Linux
MacOS
Android™
iPhone®
PS3
Nintendo 3DS™
Wii
Wii U™

make Scaleform make Linux
cygwin
Windows "Cygwin Bash Shell" "ld" "make"

1. setup.exe <http://www.cygwin.com/>
2. 'Install from Internet'
3. (c:/cygwin)

4. . (c:/cygwin)

5. 'Internet Connection' 'Direct Connection' .

6. , .

7. 'Select Packages' 'Devel' , 'binutils' 'make' .

8. .

, "Cygwin Bash Shell" "ld" "make" .

```
$ make -v
```

```
GNU Make 3.81
```

```
Copyright (C) 2006 Free Software Foundation, Inc.
```

```
...
```

```
This program built for i686-pc-cygwin
```

```
$ ld -v
```

```
GNU ld (GNU Binutils) 2.18.50.20080625
```

```
===
```

Scaleform .

make .

1. Scaleform Makeconfig .(
.) .

a. export CELL_SDK := <path to cell sdk>

b. EPATH := \$(PATH)

c. export PATH = \$(CELL_SDK)/host-win32/ppu/bin:\$(CELL_SDK)/host-
i. win32/Cg/bin:\$(CELL_SDK)/host-win32/bin:\$(EPATH)

2. Makeconfig

. Use /<drive letter>/... (msys) /cygdrive/<drive
letter>/... (Cygwin) , CELL_SDK C:\ps3\cell

a. "export CELL_SDK := /cygdrive/c/ps3/cell".

3.

a. \$ make P=PS3

b. \$ make P=PS3 C=Debug

```
make          C=<config>          .          "+"
          .          , RTTI          "make C=Release+NoRTTI"
          .          Makefile          ,          Projects
          makefile          .
```

2.2.9.2 'make' 사용 예:

공통 사용:

```
make P=Android C=Debug
make P=3DS C=Debug+NoRTTI
```

SNC 컴파일러를 사용하는 PS3 용 빌드

```
make P=PS3+snc
```

GCC 를 사용하는 PS3 용 빌드

```
make P=PS3
```

non-SCU lumping 을 사용하는 iOS 용 빌드(SCU 빌드의 다음 섹션 참조)

```
make P=iPhone/armv7 C=Release
```

자세한 출력과 함께 빌드:

```
make P=wii C=Shipping+NoRTTI VERBOSE=1
```

Mac 용 RTTI 와 함께 x64 립 릴리스 빌드:

```
make P=MacOS/x86_64 C=Release
```

iPhone 시뮬레이터 또는 armv7 용 빌드:

```
make P=iPhone/sim
make P=iPhone/armv7 C=Release
```

단일 대상 빌드(립 또는 실행 파일):

```
make P=MacOS C=Release+NoRTTI Lib/MacOS-x86_64/Release_NoRTTI/libgfx.a
```

파일 4 개를 동시에 컴파일(-j4):

```
make P=PS3 C=Shipping+NoRTTI -j4 Bin/PS3/FxPlayer_Shipping_NoRTTI.elf
```

2.2.9.3 SCU 빌드

Scaleform (SCU)

Lumping Unity SCU

Scaleform SDK

SCU

SCU

Scaleform

SCU

AS3 Obj Accessibility SCU

Src/GFx/AS3/Obj/Accessibility/AS3_Obj_Accessibility_All.cpp

```
...
#include "AS3_Obj_Accessibility_ISearchableText.cpp"
#include "AS3_Obj_Accessibility_ISimpleTextSelection.cpp"
#include "AS3_Obj_Accessibility_Accessibility.cpp"
#include "AS3_Obj_Accessibility_AccessibilityImplementation.cpp"
#include "AS3_Obj_Accessibility_AccessibilityProperties.cpp"
```

SCU

Visual Studio SCU

SCU :

AS2_SCU

AS3_SCU

AIR_SCU

Non-SCU :

AS2

AS3

AIR

non-SCU

make SCU

SCU

make

'SCU=0'

make P=PS3+snc C=Release+NoRTTI SCU=0

2.3 Windows 용 Scaleform SDK

Scaleform on Windows® XP (32 bit, SP3), Vista (32/64 bit, SP2) Windows 7 (32/64 bit) . Visual Studio 2009 (Visual Studio 2010) .

2.3.1.1 Scaleform 라이브러리 구성

Scaleform Scaleform .

Scaleform SDK .

WProgram FilesWScaleformWGfx SDK 4.3WLibW[Platform]W[MSVC Version]W

4

Debug	Scaleform Debug libs
Runtime Library	Debug (/MTd)
Debug Information Format	C7 (/Z7)

DebugOpt	Scaleform Debug
Runtime Library	(/MT)
Debug Information Format	C7 (/Z7)
Optimization	(/Ox)

Release	Scaleform Release
---------	--

Runtime Library	(/MT)
Optimization	(/Ox)

Shipping	Scaleform	Shipping libs
Runtime Library	(/MT)	
Optimization	(/Ox)	

2.3.1.2 MSVC 프로젝트 설정

Scaleform (SWFToTexture, TextureInSWF 등) "Working
Directory" . C:\Program Files (x86)\Scaleform\GFx SDK
4.3\Bin\Data

1. Project .
2. Properties() .
3. Configuration All Configurations .
4. Debugging .
5. Working Directory .

2.4 Mac 용 Scaleform SDK

Scaleform SDK Scaleform Cocoa GL MacOS

.

2.4.1 요구 사항

Scaleform

.

MacOS Snow Leopard(10.6)/Lion(10.7)

Mac/iPhone/iPad Xcode(3.2.6)

: PPC

.

2.4.2 설치

MacOS Scaleform

.

MacOS

.

.

sf_4.3_eval_macos_i686.tar.bz2

sf_4.3_eval_macos_x86_64.tar.bz2

src

.

sf_4.3_src_macos_i686.tar.bz2

sf_4.3_src_macos_x86_64.tar.bz2

()

src

.

sf_4.3_lib_macos_i686.tar.bz2

sf_4.3_lib_macos_x86_64.tar.bz2

o sf_4.3_{eval/lib/src}_video_macos_i686.tar.bz2

2.4.3 MacOS X(10.6 및 10.7)용 구성

Scaleform MacOS SDK terminal.app Scaleform .

x86 x64 Scaleform() :

\$ make P=MacOS

Bin/MacOS .

cpu Scaleform() :

\$ make P=MacOS/i686

\$ make P=MacOS/x86_64

Zlib .

2.4.4 실행 파일 실행

Bin/MacOS-i686 Bin/MacOS-x86_64 .

Finder .

: \$ *Bin/MacOS-i686/FxPlayer.app/Contents/MacOS/FxPlayer <swf file>*

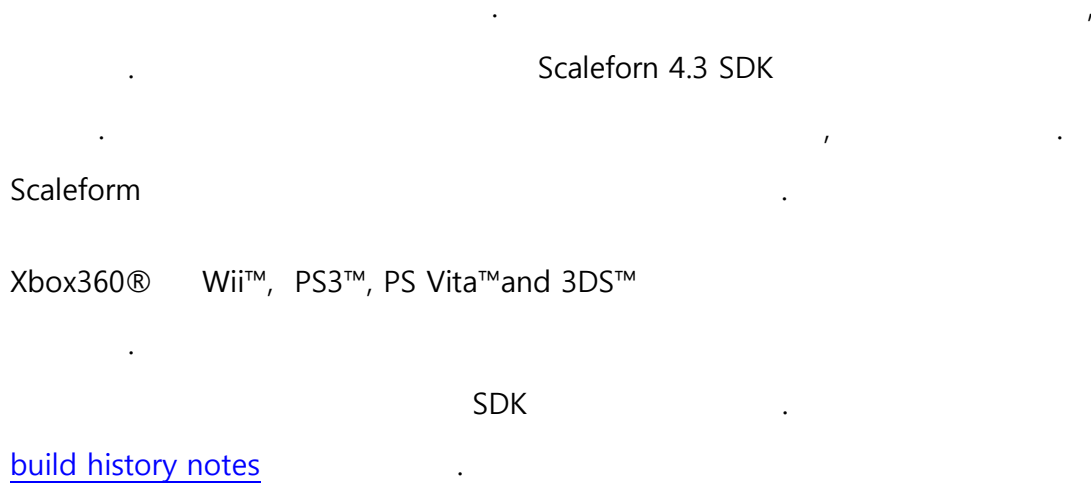
Finder SWF Dock , . HUD

AS2 AS3 Finder(*Bin#Data#AS3#Kits#Menu Bin#Data#AS3#Kits#HUD*)

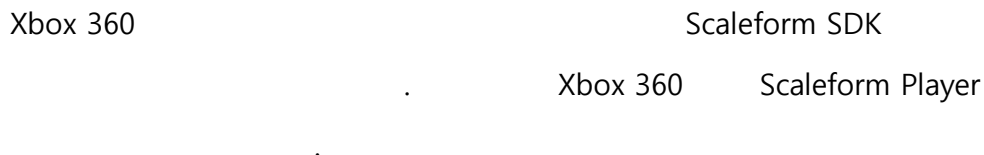
.

(): *\$Bin/MacOS-i686/MenuDemo.app/Contents/MacOS/MenuDemo*

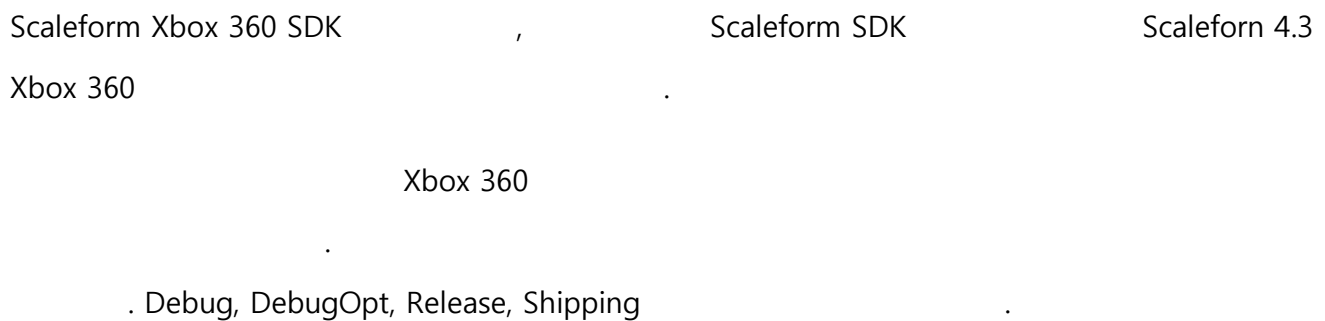
2.5 콘솔용 Scaleform SDK



2.5.1 Xbox 360 을 위한 Scaleform SDK



2.5.1.1 플레이어 및 예제들에 대한 빌드 및 구동



```

Scaleform
"Build"
"Build Solution"
( , Scaleform SWF ) Xbox
Xbox
Xbox
"
"
"
"
. (
.) Xbox
Xbox
Xbox
"
"
"
"
. (
.) Xbox
F5
. Scaleform 4.2 Xbox 360
Player, SWFTToTexture, TextureInSWF, PlayerTiny
Xbox
, Xbox
Xbox
A
Xbox
Player, SWFTToTexture, TextureInSWF,
PlayerTiny
(GFx_Xbox360,
GFxExpat, Sound)
Xbox 360®
FxPlayer
focus
(
)
A
focus
(Enter)
Y
X
B

```

RB

LB

Start /

Back .

<back> , FxPlayer Flash

.
.

A : Enter,

B : Escape,

Start : ~,

LT : PgUp,

RT : PgDn,

LB: Home,

RB : End,

LTH : Insert

RTH : Delete

FxPlayerXbox360.cpp FXPLAYER_FILENAME Flash (SWF)

.

:

#define FXPLAYER_FILENAME "Window.swf"

"Window.swf" focus A .

2.5.1.2

Scaleform (, Xbox360), Scaleform SDK .

Xbox360 Scaleform .

1. . Scaleform
 . {GFX_SDK}/Projects/Xbox360/Msvc90/SDK/GFx 4.3 Xbox 360 3rd Party.sln
, (Release) .

2. Open {GFX_SDK}/Projects/Xbox360/Msvc90/SDK/GFx 4.3 Xbox 360 SDK.sln .

3. .

:

RTTI() C++ Xbox360

.

2.5.2 PS3 를 위한 Scaleform SDK

PS3 Scaleform SDK SDK readme

.

2.5.2.1 시스템 요구 사항

Cell SDK 250 FMOD . GNU

3.81 . ()

ProDG . PS3 SNC

.

libgfxrender_ps3.a .

,

,

libgcm

Lib/PS3/Shaders.a

PSGL

Lib/PS3/GLShaders.a

make

2.5.2.2

Lib/<platform>/libgfxrender_<renderer>.a

2.5.2.3 make 사용

Make Scaleform

1. Sec 2.2.9.1 'Make Scaleform'

2. Scaleform Makeconfig .(
.)

```
export CELL_SDK := <path to cell sdk>  
EPATH := $(PATH)  
export PATH = $(CELL_SDK)/host-win32/ppu/bin:$(CELL_SDK)/host-  
win32/Cg/bin:$(CELL_SDK)/host-win32/bin:$(EPATH)
```

Makeconfig

. Use /<drive letter>/... (msys) /cygdrive/<drive

letter>/... (Cygwin) , CELL_SDK C:\ps3\cell

"export CELL_SDK := /cygdrive/c/ps3/cell".

3. Scaleform

\$ make P=PS3

SNC :

\$ make P=PS3+snc

, "normal" PS3 snc ,
snc snc . (make P=PS3+snc
.)

4. \$(CELL_SDK)/FxPlayer/flash.swf SWF . SWF

Scaleform L1 R1 ScaleformPlayer .

5. "\$\$(CELL_SDK)/FxPlayer/sf_license.txtsf_license.txt"

.(cell .) RTC PRX .

cellSysmoduleLoadModule(CELL_SYSMODULE_RTC);

6. ProDG . FxPlayer

\$(CELL_SDK)/FxPlayer .

7. SWFTToTexture RenderTexture ,

Bin/Data/AS2/Samples/SWFTToTexture Bin/Data/AS2/Samples/RenderTexture

SWF \$(CELL_SDK)/FxPlayer .

2.5.2.4 Visual Studio 사용

Make , Scaleform

.(PS3)

Scaleform , PS3

Projects/PS3/Msvc90/SDK Scaleform 4.3 PS3 SDK.sln

. Projects/PS3/Msvc90/Demos

Scaleform 4.3 PS3 Demos.sln

2.5.2.5 플레이어 제어

"PS3" Scaleform

- focus ()
- focus (Enter)
-
- HUD
-
- R1, L1 ,
- L2
- R2 , /
- Start /
- Select

<select> , FxPlayer FxPlayerAppBase
ctor(PadKeyCommandMap) Flash

```

: "Window.swf" focus (X)
. FxPlayer "Window.swf" /usr/local/cell/FxPlayer/flash.swf
. SWF/GFX L1/L2
.

```

2.5.2.6 PS3 렌더러 지원

, GCM GL PS3 .

libgcm:

```

: Src/Render/PS3/
.

```

```

Lib/PS3/Shaders.a , .

```

```

$ make P=PS3 Lib/PS3/Shaders.a

```

PSGL:

```

Lib/PS3/GLShaders.a , .

```

```

$ make P=PS3 Lib/PS3/GLShaders.a

```

```

GLShader.a SN . GNU "ppu-lv2-g++ -mno-sn-

```

```
ld" .

```

```

libcgm PSGL Scaleform player .

```

libgcm :

```

PlayerGCM SWF

```

```

PlayerTiny SWF

```

```

SWFToTextureGCM SWF

```

```

TextureInSWFGCM SWF

```



```

PSGL      :
    PlayerGL      SWF
    PlayerTinyGL  SWF
    SWFToTextureGL SWF
    TextureInSWFGL SWF
    ,
    PSGL

```

libgcm .

```

PSGL  Apps/Samples/Common/OpenGLPS3App.*
    SWFToTexture/TextureInSWF  Apps/Samples/Common/MathLib.*
    .
    .

```

```

#ifdef FXPLAYER_RENDER_GCM
    <gcm specific>
#else
    <psgl specific>
#endif

```

2.5.2.7 빌드 구성

.(Release)

```

Shipping .
Release .
Debug
DebugOpt
NoRTTI RTTI .
NoThreads Scaleform .(SF_ENABLE_THREADS)
    C=<config> . + .
    , "make C=Release+NoThreads" .

```

2.5.2.8 통합에 있어서의 주의사항

Scaleform PRX . (resc, fs, gcm)

```
cellSysmoduleLoadModule(CELL_SYSMODULE_RESC);  
cellSysmoduleLoadModule(CELL_SYSMODULE_FS);  
cellSysmoduleLoadModule(CELL_SYSMODULE_GCM);
```

"Date" () rtc .

```
cellSysmoduleLoadModule(CELL_SYSMODULE_RTC);
```

2.5.2.9

SF_ENABLE_THREADS

. SF_ThreadsPthead.cpp

2.5.3 PS Vita 용 Scaleform SDK

2.5.3.1 요구 사항

Vita SDK

2.5.3.2 설치

PS Vita Scaleform SDK

PS Vita

sf_4.3_psvita_vc90_eval.exe

src

sf_4.3_psvita_vc90_src.exe

()

sf_4.3_psvita_vc90_lib.exe

2.5.3.3 플레이어/샘플 구성

Scaleform PS Vita SDK

Scaleform 4.3 PS Vita Demos Visual Studio

. "Projects/PSVita/Msvc90/Demos/GFx 4.3 PSVitaDemos.sln"

FxPlayerMobile FxPlayerTiny Release, Debug, DebugOpt Shipping

. PS Vita Scaleform Scaleform SDK

PS Vita Scaleform

1. "Projects/PSVita/Msvc90/SDK/GFx 4.3 PSVita SDK.sln"

2. Build Solution

make Vita

2.2.9 'make

Scaleform

2.5.3.4 플레이어 및 샘플 실행

FxPlayerTiny Scaleform

FxPlayerMobile HUD

 (SWF) Vita "File Serving Directory" "FxPlayer" .
FxPlayerMobile "flash.swf" SWF
"flash.swf" .

2.5.3.5 평가 버전

2.10 .

2.5.4 Wii 를 위한 Scaleform SDK

Wii Scaleform SDK read me (readme_wii.txt, readme_make.txt)

2.5.4.1 플레이어 및 샘플 실행하기

Bin/wii Codewarrior ndrun . Codewarrior

, . F5

. (.)

DVD . RVL_SDK/dvddata/FxPlayer

dvddata/FxPlayer/gfxlicense/txt .

.

Player flash.swf (SWF GFX)

PlayerTiny Window.swf (SWF GFX)

```
TextureInSWF      Bin/Data/AS2/Samples/RenderTexture
                  dvddata/FxPlayer
                  .
```

- and + (dvddata/FxPlayer SWF GFX)
C
Z .

```
<Z>          ,          FxPlayer          FxPlayerAppBase
ctor(PadKeyCommandMap      )          Flash
```

2.5.4.2 샘플 빌드하기

Scaleform

Shipping

Release

Debug

DebugOpt

+NoRTTI

RTTI

+Sdata0

No small data section (-sdata 0 -sdata2 0)

+NoThreads

Scaleform

(SF_ENABLE_THREADS,

.)

.Wii

+NoRTTI

RTTI

+ _

Codewarrior
projects"

, Preference

IDE Extras group

"Use text-based

XML

. Codewarrior

Make

libgfx_video.a

C/C++

"#define GFX_ENABLE_VIDEO"

Scaleform

. (Cygwin

. PS3

2.3.2.3

Scaleform

Cygwin

.)

```

, Wii SDK          SDK          Makeconfig          .(
.)

```

```

WII_CWDIR := "/cygdrive/c/Program Files/Freescale/CW for Wii v1.0"
WII_SDKDIR := /cygdrive/c/RVL_SDK
WII_NDEVDIR := /cygdrive/c/NDEV

```

```

Makeconfig          ,          ,          .
/cygdrive/<drive letter>/...          .
.

```

```

cygwin          "make P=wii C=DebugOpt+NoRTTI"          .
DebugOpt+NoRTTI          .
.          C          . Makeconfig
RVL_NDEV          .

```

2.5.5 Nintendo 3DS 용 Scaleform SDK

2.5.5.1 요구 사항

```

CTR SDK          .

```

2.5.5.2 설치

```

3DS          Scaleform SDK          .
3DS          .
sf_4.3_eval_3ds.tar.bz2
src          .
sf_4.3_src_3ds.tar.bz2

```

()
sf_4.3_lib_3ds.tar.bz2

2.5.6.3 코드 구성

Scaleform 3DS SDK Cygwin Scaleform .

Scaleform() :

\$ make P=3DS

Bin/3DS .

2.2.9 'make Scaleform ' .

2.5.6.4 플레이어 및 샘플 실행

FxPlayerMobile / FxPlayerTiny Projects/3DS/Makefile, STARTUP_FLASH

FxPlayer_CONTENT Flash .

. SWF Obj/3DS/rom_FxPlayerMobile/FxPlayer cci

.

CTR .

2.5.6.5 평가 버전

2.10 .

2.6 모바일용 Scaleform SDK

Scaleform

Scaleform

iOS

2.6.1 Android 용 Scaleform SDK

2.6.1.1 요구 사항

Scaleform

[Android SDK](#)

[Android NDK](#)

Android Platform API

2.6.1.2 설치

Android Scaleform SDK

Android

sf_4.3_eval_android.tar.gz

src

sf_4.3_src_android.tar.gz

()

sf_4.3_lib_android.tar.gz

2.6.1.3 플레이어/샘플 빌드 및 실행

1. Scaleform Makeconfig

```
export JAVA_HOME := <JDK >
ANDROID_SDK := <SDK >
ANDROID_NDK := <NDK >
export PATH := $(JAVA_HOME)/bin:$(ANDROID_SDK)/tools:$(ANDROID_SDK)/platform-
tools
```

```
Makeconfig (slash) UNIX
.msys /< >/... , Cygwin /cygdrive/<
>/... , Android_SDK C:\Android\SDK
```

```
"export ANDROID_SDK := /cygdrive/c/Android/SDK".
```

2. Scaleform

```
$ make P=Android
```

3. Bin/Android APK . APK

4. SWF /sdcard/flash.swf . SWF GFx HUD Scaleform Player .

5. "/sdcard/sf_license.txtsf_license.txt" .

```
flash.swf
"/sdcard"
```

2.6.1.4 소스 빌드하기

•

 Springer

•

•

•

•

•

•

.

iOS . Xcode .

Debug, DebugOpt, Release Shipping .

Scaleform . Xcode "Build and Run" .

(GFX SWF)

iOS .

iOS "flash.swf" SWF .

SWF iTunes

iTunes Apps "File Sharing" , Apps FxPlayerMobile SWF "FxPlayerMobile Documents" "Add" SWF .

2.6.2.4 소스 구성

Scaleform Scaleform SDK . iOS

Scaleform .

1. Scaleform iOS . lipo (Release) make (:

make P=iPhone/armv7 C=Release

2.6.3 모바일 플레이어에서의 수명 주기 관리와 관련하여 자주 묻는 질문(FAQ)

Scaleform SDK for Mobiles ,

2.7 CS3 용 Scaleform 런처 패널 설치하기

Scaleform

SWF

Scaleform player

1. (→ → Adobe Extension Manager CS3) C:/Program Files/Adobe/Adobe Extension Manager.

Finder>Applications>Adobe Extension Manager

CS3>Extension Manager

2. ,

3. : C:/Program Files/Scaleform/GFx SDK 4.3/Resources/Tools/
: scaleform_gfx_4.3_macos/Resources/Tools/

4. Scaleform Extensions.mxp

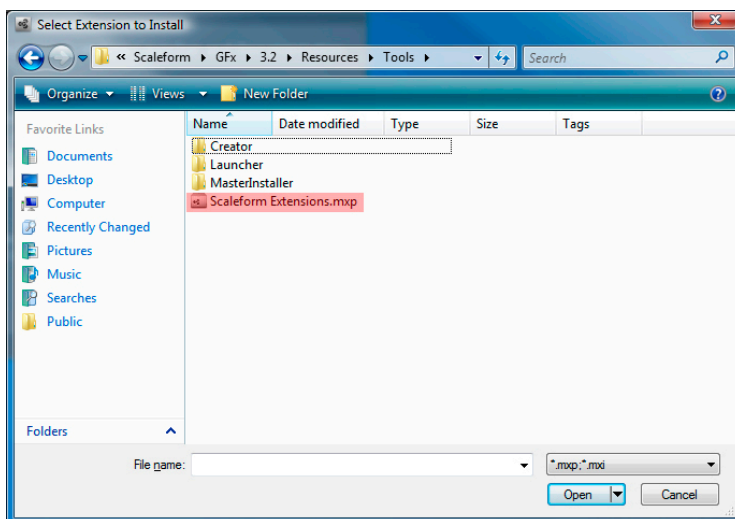


그림 1: Scaleform Extensions.mxp 파일 위치 (Windows)

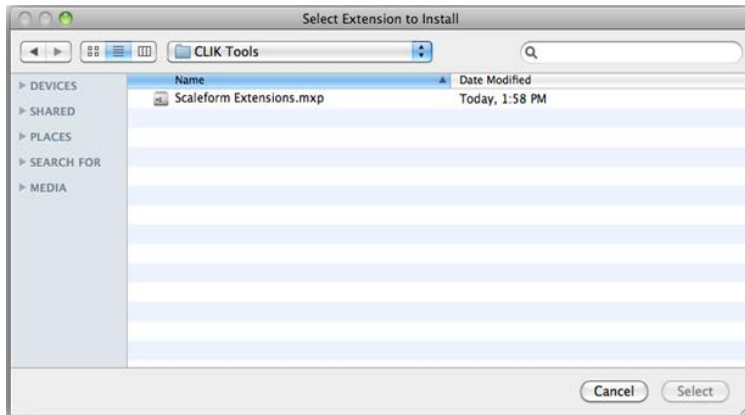


그림 2: Scaleform Extensions.mxp 파일 위치 (Mac)

5. *Accept*

6. OK

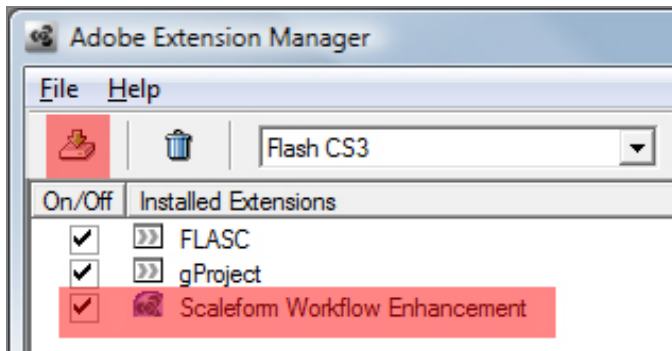


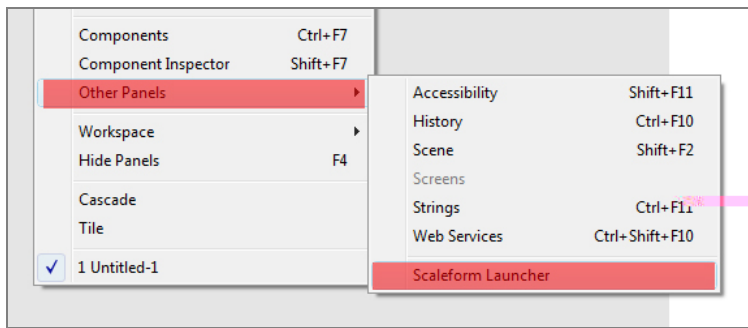
그림 3: CS3의 어도브 확장 매니저 (Windows)

그림 4 CS3의 어도브 확장 매니저 (Mac)

7.

8. Flash , Flash (2.0 or 3.0)

9. Flash *Other Panels* Scaleform Launcher .



4. *Scaleform Extensions.mxp*

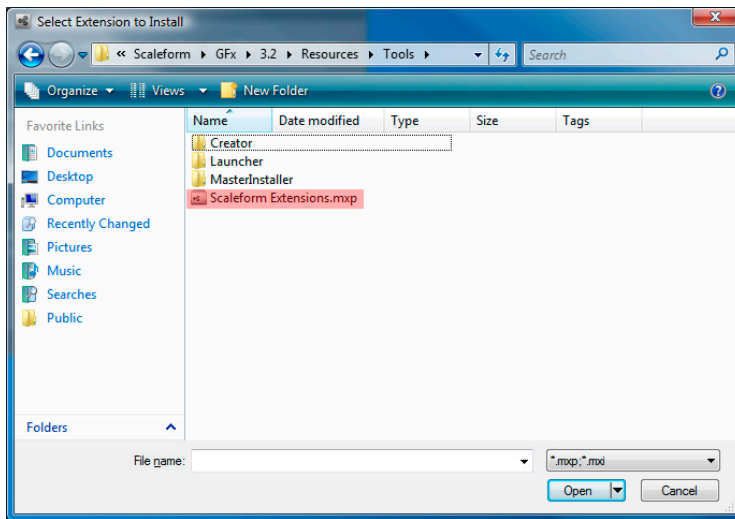


그림 7: Scaleform Extensions.mxp 파일 위치 (Windows)

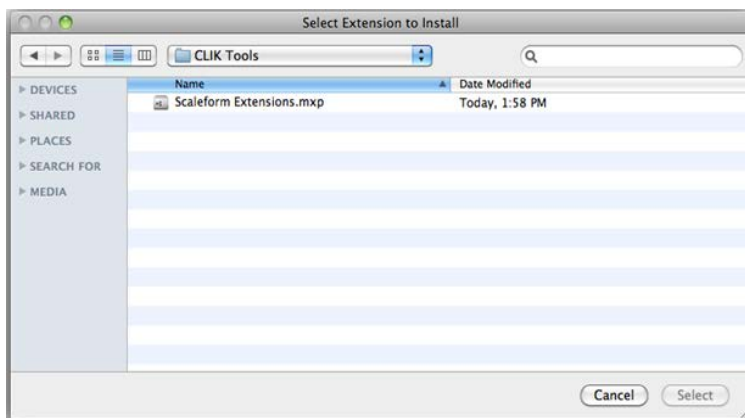
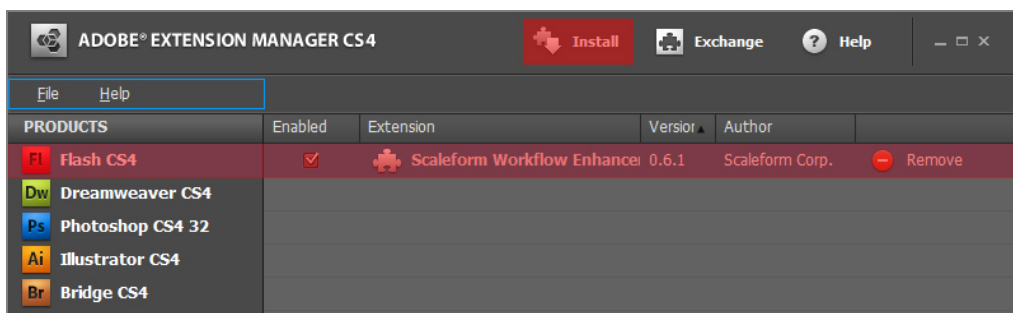


그림 8: Scaleform Extensions.mxp 파일 위치 (Mac)

5. *Accept*

6. *OK* CS4 or CS5



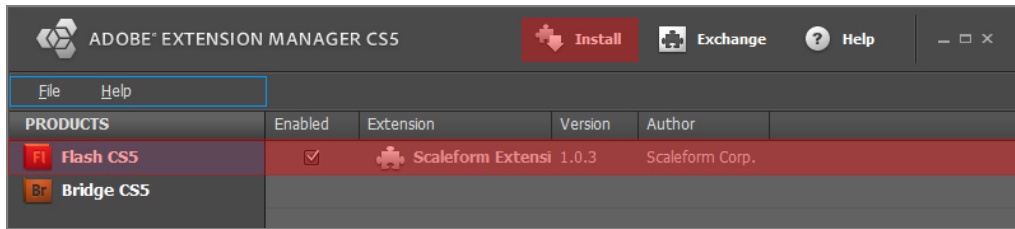


그림 9: CS4(top) or CS5 (bottom)의

2.9 Scaleform 런처 패널 사용하기

Scaleform	Scaleform		
Scaleform	SWF	.	CLIK
Scaleform		Scaleform	.

. [+]

Add New Profile

그림 13: [+]버튼을 눌러서 프로파일 추가 하기

주의:

NO

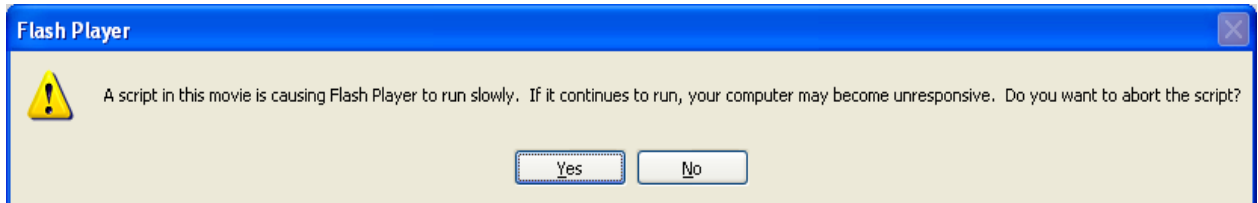


그림 14: 스크립트 취소 대화상자

2.

[+]

Scaleform

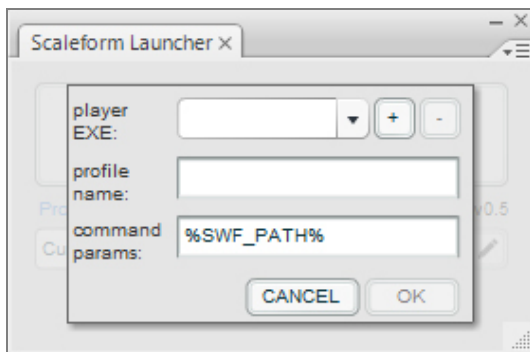


그림 15: Add new profile 대화상자.

3.

GFxMediaPlayer

.

Scaleform

. GFxMediaPlayer

. C:/Program Files/Scaleform/GFx SDK 4.3/Bin

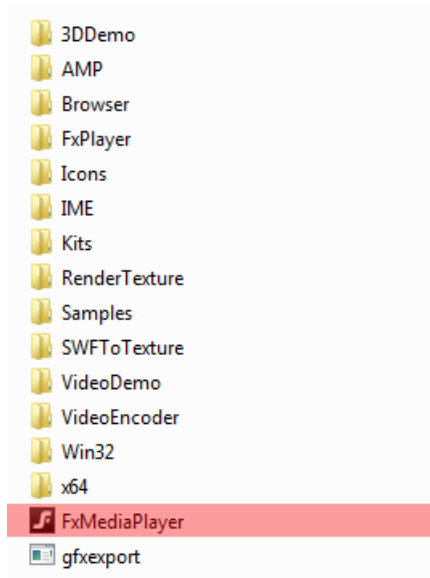


그림 16: GfxMediaPlayer 실행파일 찾기(Windows)

scaleform_gfx_4.3_macos/Bin/MacOS/FxPlayer_DebugOpt.app/Contents/MacOS
GfxPlayer_DebugOpt .

그림 17: GfxPlayer_DebugOpt 실행파일 찾기 (Mac)

4.

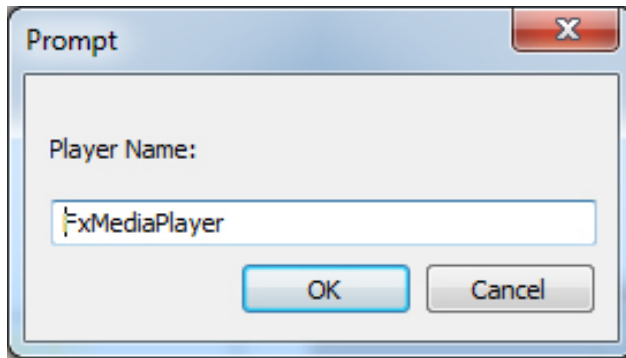


그림 18: 플레이어명 결정

그림 19: 플레이어명 결정

5. OK .
6. 'DebugOpt' Add Profile .
AMP .
7. .

'%SWF_PATH%' , .

사용예: '%SWF_PATH% -nh'

HUD SWF GFx .

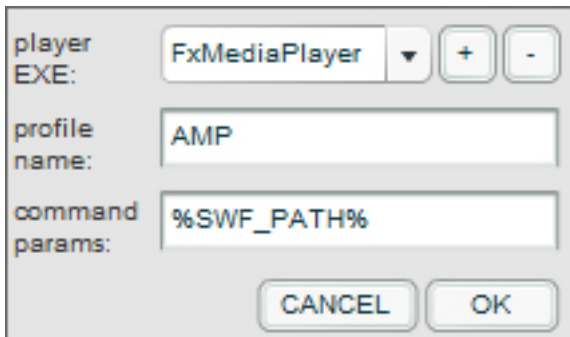


그림 20: 기본 전달인자 상태인 AMP 프로파일 (windows)

그림 21: 기본 전달인자 상태인 GfxPlayer_DebugOpt 프로파일 (Mac)

8. OK .
9. *Profiles Scaleform*

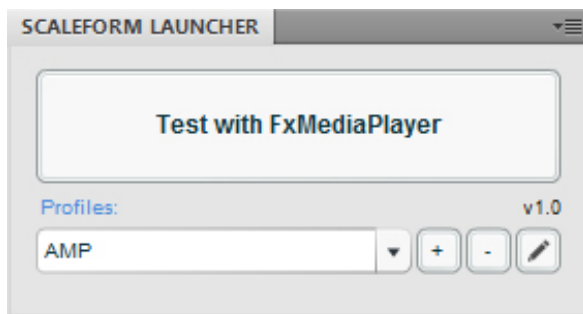


그림 22: AMP 프로파일이 등록된 Scaleform 런처 패널

그림 23: Debug_Opt 프로파일이 등록된 Scaleform 런처 패널

프로파일 수정

1. .
2. *Edit*

3. , , .
4. OK

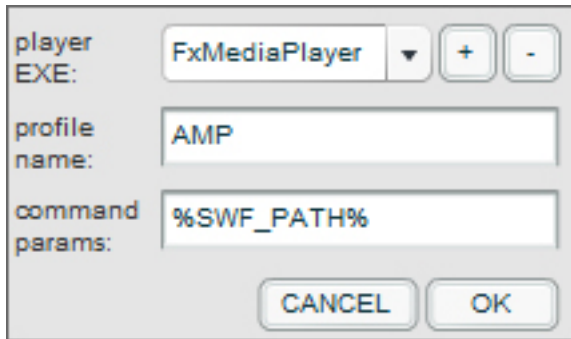


그림 24: 프로파일 수정/삭제 대화상자 (Windows)

그림 25: 프로파일 수정/삭제 대화상자 (Mac)

프로파일 삭제

1. Profile
2. [-] .
3. OK . Cancel

SWF 테스트

1. SWF .
2. .
3. Test with [] .

4. Scaleform player movie .

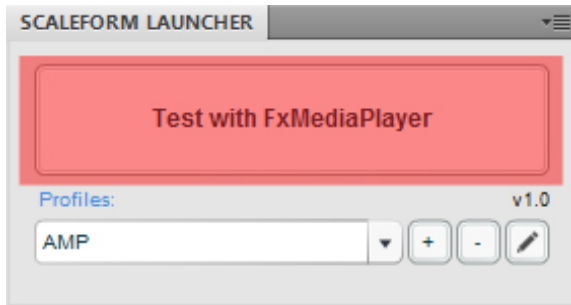


그림 26: Test with GfxMediaPlayer 버튼 누르기(Windows)

그림 27: Test with GfxPlayer_DebugOpt 버튼 누르기(Mac)

키보드 바로 가기로 플레이어 실행하기

Flash Scaleform Player .

1. *Edit* .
2. *Keyboard Shortcuts* .
3. Adobe Standard Current . (Keyboard Shortcuts)
4. *Duplicate Set* .
5. () .

6. *OK* .
7. "*Commands* Drawing Menu Commands" .
8. *Commands* + .
9. Scaleform Player Publish .
10. + .
11. Press (Ctrl+Enter) Ctrl+Enter .
Flash .
12. *Change* .
13. , *Reassign* .
14. *Commands* Control .
15. Test Movie .. Flash publish .
16. + .
17. Press (Ctrl+\ (backslash)) .
18. *Change* .
19. *OK* Keyboard Shortcuts .

2.10

2.10.1 라이선스 문제

Scaleform 4.3

sf_license.txt

. sf_license.txt

3AAAA1BB23B8ZCCCC4CCEOFSJCZ08DDDDDEF

sf_license.txt

. Scaleform 4.3

SDK

sf_license.txt

sf_license.txt

sf_license.txt

Scaleform 4.3

The Scaleform trial period expired on 1(m) 1(d) 2008, please contact Scaleform for an extension. Error: 106

sf_license.txt

Failed to open the Scaleform license file, please make sure sf_license.txt exists in the working directory. Error: 101

Windows / Mac® / Linux®

1.

2. GFXSDK

3. GFXSDK , HOME

Xbox 360®

1. XDK Debug Drive - D:/\$ (SolutionName)/sf_license.txt

.

“PSP®” (PlayStation®Portable)”

1. :
Linux - /usr/local/psp/FxPlayer/sf_license.txt
Windows - C:/usr/local/psp/FxPlayer/sf_license.txt

“PLAYSTATION®3”(PS3™)

1. /app_home/sf_license.txt
2. /host_root/usr/local/cell/FxPlayer/sf_license.txt
3. /dev_bdvd/PS3_GAME/USRDIR/sf_license.txt

PLAYSTATION®Vita”(PS Vita™)

1. "FxPlayer" sf_license.txt

.

3DS™

1. sf_license.txt Lib/3DS/target .

Wii™

1. dvddata/FxPlayer/sf_license.txt

iOS

1. sf_license.txt Xcode Xcode (:
Scaleform.app) .

Android

1. sf_license.txt / "sdcard"
(root
).

2.11

Scaleform

웹사이트:

Scaleform

<http://gameware.autodesk.com/scaleform>

<http://gameware.autodesk.com/scaleform/developer/>

Scaleform

, FAQ

티켓 시스템:

, e-mail

. e-mail

, ()

Scaleform

포럼:

Scaleform

Scaleform

Scaleform

e-mail

2.12

Scaleform

gameware@autodesk.com

Scaleform

3 동봉된 것들

Scaleform 4.3 . Scaleform 4.3
 , Scaleform 4.3 , Scaleform 4.3 ,
 Scaleform 4.3 Flash AS 2.0 .
 , , , , ,
 , , , , , .
 Scaleform 4.3 , Doc
 .
 .
 1. SDK
 2. : → → Scaleform → GfX SDK 4.3 → Documentation

3.1 Flash

Flash AS2 . Flash
 Scaleform IGDA (International Game Developers Association) Flash SIG (Special
 Interest Group) http://www.igda.org/wiki/Flash_SIG .
 Flash , IGDA SIG
 .
 Flash .
[Lynda.com](http://www.lynda.com)
[Essential ActionScript 2.0 by Collin Mook](#)
[ActionScript: The Definitive Guide by Collin Mook](#)

3.2

Scaleform 4.3

Scaleform 4.3

Getting Started with CLIK: CLIK Flash, CLIK, ,

Getting Started with CLIK Buttons: CLIK

Getting Started with CLIK

Getting Started with Video: , USM video

exporting, Video Flash Video

: Scaleform Video

Getting Started with IME: IME

3.3

Scaleform 4.3 Flash

Integration Tutorial: Scaleform 4.3 3D DirectX 9

Reference Documentation: CHM , Scaleform

4.2

DrawText API: Scaleform 4.3 C++

DrawText API

ActionScript Extensions: Scaleform 4.3

Font Overview Configuration: Scaleform 4.3

Scaleform C++

API

Scale9Grid: Scaleform 4.3

Scale9Grid

XML Overview: Scaleform 4.3 XML

Flash Support Overview: Scaleform 4.3

(XLS)

Flash

Memory System Overview: Scaleform 4.3

AMP,

IME Configuration: Scaleform 4.3 IME

IME

3.4

, UE3

SDK

gameware@autodesk.com

IME Xbox 360 Configuration: Xbox 360 IME

IME PS3 Configuration: PS3 IME

Unreal Engine 3 Workflow: Flash

3

Unreal Engine 3 Integration:

Scaleform 4.3 UE3

. Scaleform 4.3 Kismet, ,

UE3 .

README -

readme . readme SDK .

3.5

Scaleform 4.3 . Scaleform

. Demos

"Downloads" .

<http://gameware.autodesk.com/scaleform/developer/?action=demo>

3D

.

Bin/<platform>/ directory

4 어디서 시작할 것인가

Scaleform 4.3

Player

Scaleform 3.3

Scaleform 4.3

Scaleform

Scaleform 4.3

Scaleform 4.3

Scaleform 4.3

UI /

1

UI 1

Flash

Flash

). FAQ

. <http://gameware.autodesk.com/developer/forums/>

4.1

4.1.1 UI 프로그래머

Scaleform

DB

2D Flash

Scaleform 4.3

DB () Flash

Flash

(: , , HP).

4.1.2 UI 디자이너

Flash , AS2
CLIK
/

4.1.3 UI 아티스트

CS3 CS4, Flash
CLIK)
UI

Scaleform 4.3 Flash

4.2

Scaleform 4.3 . Scaleform 4.3
SWF Scaleform Scaleform
4.3 player SWF GFX . Scaleform

Windows Scaleform 4.3 SDK "Scaleform SDK Browser"

 . SDK Browser 'Programs' Scaleform

Player .

→ → Scaleform → GfX SDK 4.3 → GfX Players

C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Win32/Msvc80/GfXPlayer

C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Win32/Msvc90/GfXPlayer

SDK Browser Scaleform GfX Player .

C:/Program Files/Scaleform/GfX SDK 4.3/Bin/Data/AS2/Samples/3DDemo/ 3D

SWF .

Bin/<platform> . Scaleform SDK 4.3 Finder()

SWF , Scaleform 4.3 Bin 3DDemo SWF

SWF . SWF

- Bin/i386-linux/FxPlayer <swf file>

OS - Bin/MacOs/FxPlayer.app/Contents/MacOS/FxPlayer <swf file>

: Scaleform SDK 4.3 */Bin/Data/AS2/Samples*

4.3

Scaleform 4.3 AS 2.0, 3.0 . Flash CS3, CS4 CS5

. , Flash AS

2.0 AS3.0 . Flash File → New → Flash File

(ActionScript 2.0 ActionScript 3.0) .

Scaleform 4.3 AS 2.0, 3.0 .

sf_4.3_flash_support.pdf AS 2.0 .